

Enabling SME Participation in Agricultural Futures: A Tokenized Pooling Model for Euronext Wheat Contracts

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Abstract — Small and medium-sized enterprises (SMEs) in agriculture face significant barriers to participating in institutional commodity trading, limiting their ability to manage risk and access global markets. This paper introduces a novel tokenized pooling mechanism that allows SMEs to collectively access Euronext futures trading for agricultural commodities. Leveraging blockchain technology, specifically Hyperledger Fabric, we create a transparent and efficient system for aggregating small-volume trades into standard-size contracts. Our model incorporates advanced pricing techniques that account for collateralization and cross-currency considerations. Implementation on Hyperledger Fabric demonstrates the feasibility of this approach in a real-world setting. Results indicate that this pooling mechanism can significantly reduce counterparty risk for SMEs while providing them access to institutional-grade risk management tools. This innovation has the potential to enhance market efficiency and promote more inclusive global agricultural trade.

Keywords — agriculture, tokenization, commodity futures, Euronext futures trading, Hyperledger Fabric

I. INTRODUCTION

The global agricultural commodities market has long been dominated by large institutional players, leaving small and medium-sized enterprises (SMEs) at a significant disadvantage. These smaller producers and traders, who form the backbone of many local agricultural economies, often lack the volume and financial resources to effectively participate in institutional-level commodity trading and risk management.

Consider a small wheat farmer in Eastern Europe. These entities produce high-quality goods but in quantities too small to meet the minimum volume requirements for standard futures contracts. As a result, they're exposed to price volatilities and counterparty risks that larger players can more easily hedge against.

Our proposed solution tackles this challenge head-on through a novel tokenized pooling mechanism. Our main purpose is to allow multiple SMEs to aggregate their positions to meet the standard contract sizes required for trading on Euronext. Each participant's position is represented by tokens on a blockchain, enabling fractional ownership of standard-size Euronext futures contracts. The primary objectives of this research are threefold:

1. To develop a mathematically sound model for pricing these pooled positions, taking into account complex factors such as collateralization and cross-currency transactions.
2. To implement this model on Hyperledger Fabric, demonstrating its real-world viability and scalability.
3. To analyze the potential impact of this innovation on market accessibility, risk management, and overall efficiency in agricultural commodity trading.

Aiming to achieve these goals, we intend to democratize access to sophisticated financial instruments in the agricultural sector, potentially transforming how SMEs manage risk and engage with global markets.

II. LITERATURE REVIEW

The challenges faced by SMEs in accessing institutional commodity trading have been well-documented in the literature. Ali et al. [1] highlighted how minimum volume requirements and high collateral demands effectively shut out smaller players from institutional commodity trading. This exclusion not only impacts individual businesses but also contributes to market inefficiencies and reduced liquidity Khalatur et al. [2]. The recent years have seen growing interest in blockchain-based solutions for agricultural supply chains and commodity trading. For instance, Hasan et al. [3] demonstrated how distributed ledger technology could enhance traceability and reduce fraud in grain markets.

However, these solutions have primarily focused on supply chain management rather than financial risk mitigation. The concept of tokenization in commodity markets is not entirely new. Li [4] proposed a tokenized system for oil futures, but their model did not address the specific needs of SMEs or the aggregation of small positions. Our work builds upon these foundations while introducing novel elements to address the unique challenges faced by agricultural SMEs.

In terms of pricing models for collateralized derivatives, the seminal work of Kim [5] laid the groundwork for understanding how collateralization impacts derivative

valuation. Fujii et al. [6] extended this to multi-currency settings, which is particularly relevant for cross-border agricultural trade. Our pricing model incorporates elements from both of these approaches, adapted to the specific context of pooled agricultural futures.

The use of Hyperledger Fabric for financial applications has been explored by several researchers, including Liu et al. [7] who demonstrated its efficacy for a decentralized bond market. However, its application to agricultural derivatives, particularly in a pooled context, remains unexplored in the academic literature.

III. METHODOLOGY

Our approach combines financial modeling, blockchain implementation, and empirical analysis. The methodology can be broken down into three main components:

1. Mathematical Modeling: We developed a pricing model for pooled agricultural futures that accounts for collateralization and potential cross-currency considerations.
2. Building on the work of Menkveld [8], our model incorporates the following key elements:
 - a) Partial Differential Equation (PDE) for collateralized derivative pricing, (1):

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r_F - q)S \frac{\partial V}{\partial S} = r_F V - (r_F - r_c)C \quad (1)$$

- Where V is the derivative value;
- S is the underlying asset price;
- σ is volatility;
- r_F is the funding rate;
- q is the dividend yield;
- r_c is the collateral rate;
- and C is the collateral amount.

b) Liquidity Value Adjustment (LVA) to account for the cost of posting collateral, (2):

$$LVA = E^Q \left[\int_t^T e^{(-\int_t^s r_F(u)du)} (r_F(s) - r_c(s)) C_s ds \right] \quad (2)$$

Where:

- E^Q represents the risk-neutral expectation;
- t is the current time;
- T is the maturity time;
- $r_F(u)$ is the funding rate at time u ;
- $r_F(s)$ is the funding rate at time s ;
- $r_c(s)$ is the collateral rate at time s ;
- C_s is the collateral amount at time s ;

c) Cross-currency considerations for foreign-denominated collateral, (3).

$$V_d = E_d^Q [D_d(t, T)(S_T X_T - K_d)^+] - LVA_d \quad (3)$$

Where:

- V_d is the value of the derivative in domestic currency
- E_d^Q represents the risk-neutral expectation under the domestic measure
- $D_d(t, T)$ is the domestic discount factor from time t to T
- S_T is the price of the underlying asset at maturity
- X_T is the exchange rate at maturity (foreign currency per unit of domestic currency)
- K_d is the strike price in domestic currency
- LVA_d is the Liquidity Value Adjustment in domestic currency

3. Blockchain Implementation: The implementation on Hyperledger Fabric closely models the specifications of the Euronext Milling Wheat No.2 Futures contract (EBM) [9], with adaptations to allow for pooling of smaller positions. Key components include:

a) Asset representation: We created a digital token representing fractional ownership of a standard 50-tonne wheat futures contract. Each token can represent any quantity of wheat, allowing SMEs to participate with volumes smaller than the standard contract size.

b) Quality specifications: Our smart contract incorporates quality parameters based on the EBM contract, including moisture content, protein content, specific weight, Hagberg falling number, broken grains, and impurities.

c) Pricing mechanism: While we haven't implemented a direct pricing oracle, our contract includes a PriceFeed structure that can be updated with current market prices, allowing for accurate pricing of positions.

d) Pooling mechanism: The proposed smart contract aggregates individual token holdings into pools of 50 tonnes, the standard trading unit for the EBM contract on Euronext. Our pooling mechanism allows SMEs to initially enter positions with smaller quantities, then aggregates these positions into standardized futures contracts compatible with Euronext trading.

e) Euronext contract creation: Upon reaching the standard contract size, the model can automatically create an Euronext-compatible contract.

Figure 1 presents the class diagram of our WheatForwardToken smart contract, illustrating the key components and their relationships within our model. This diagram provides a clear overview of the smart contract structure and relationships between different components.

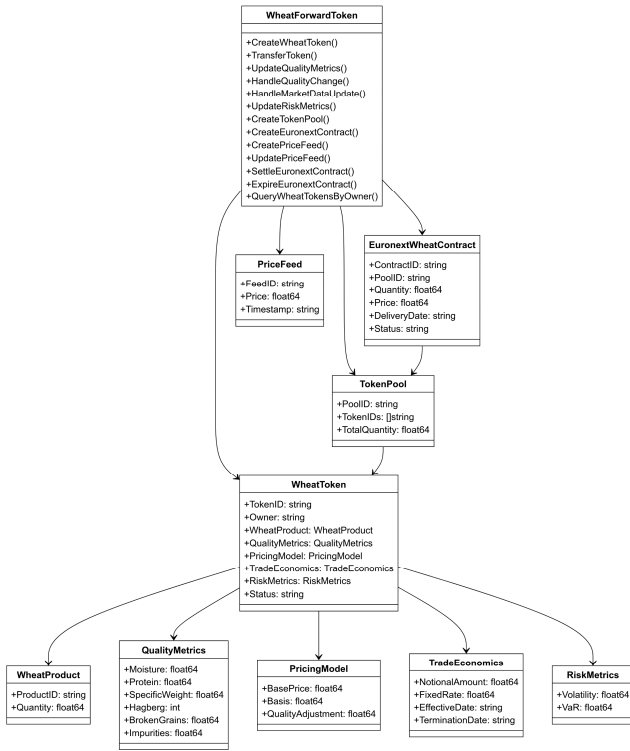


Fig. 1. Class Diagram of WheatForwardToken Smart Contract

The function described in Figure 2 demonstrates how individual wheat tokens are added to a pool, and how a Euronext contract is automatically created once the pool reaches the standard size of 50 tonnes.

```
chaincode\chaincode.go
91 // CreateWheatToken initializes a new wheat token
92 func (t *WheatForwardToken) AddToPool(ctx contractapi.TransactionContextInterface,
93     poolID string, tokenID string) error {
94     pool, err := t.GetTokenPool(ctx, poolID)
95     if err != nil {
96         return err
97     }
98
99     token, err := t.GetWheatToken(ctx, tokenID)
100    if err != nil {
101        return err
102    }
103
104    const POOL_SIZE float64 = 50.0 // 50 tonnes for Euronext contract
105
106    availableSpace := POOL_SIZE - pool.TotalQuantity
107    amountToAdd := math.Min(token.WheatProduct.Quantity, availableSpace)
108
109    pool.TokenIDs = append(pool.TokenIDs, tokenID)
110    pool.TotalQuantity += amountToAdd
111
112    if pool.TotalQuantity == POOL_SIZE {
113        // Create Euronext contract here
114        t.CreateEuronextContract(ctx, "contract-"+poolID, poolID, POOL_SIZE, 0, ""
115        // Price and delivery date to be set
116    }
117
118    poolJSON, err := json.Marshal(pool)
119    if err != nil {
120        return err
121    }
122
123    return ctx.GetStub().PutState(poolID, poolJSON)
124 }
```

Fig. 2. Core Pooling Mechanism

The sequence of operations for creating and managing wheat futures contracts in our system is depicted in Figure 3. The diagram shows the interactions between the client, peer nodes, orderer, ledger, and chaincode. The diagram illustrates the interaction between different components of the system and the flow of transactions.

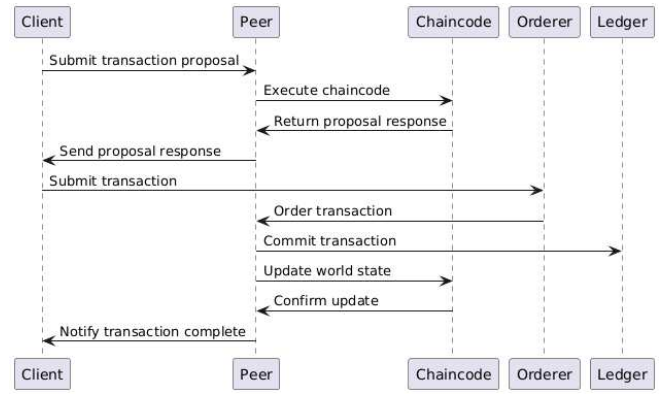


Fig. 3. Sequence Diagram of Wheat Futures Contract Creation and Management

IV. TOKENIZATION OF AGRICULTURAL ASSETS

4.1 Principles of Asset Tokenization in Agriculture

Asset tokenization in agriculture involves the creation of digital representations of physical agricultural commodities. This process allows for the fractional ownership and easier transferability of these assets, potentially increasing liquidity and market access. In our model, tokenization serves as the foundational step for enabling SME participation in agricultural futures markets.

4.2 Agrotoken Model: A Blueprint for Agricultural Asset Tokenization

Our approach draws inspiration from the Agrotoken platform [10], which has successfully implemented a tokenization model for agricultural commodities. The key components of this model include:

1. **Physical Asset Verification:** Producers deposit their agricultural products in approved storage facilities.
2. **Proof of Grain Reserve (PoGR):** A certificate is issued to verify the quantity and quality of the deposited commodities.
3. **Token Issuance:** Digital tokens are minted, representing the deposited commodities.
4. **Token Distribution:** Tokens are transferred to the producer's digital wallet.
5. **Utilization:** Tokens can be traded, used as collateral, or redeemed for physical commodities.

Figure 4 provides a detailed flowchart of the wheat tokenization process, from the initial deposit by an SME producer to the creation of Euronext-compatible contracts.

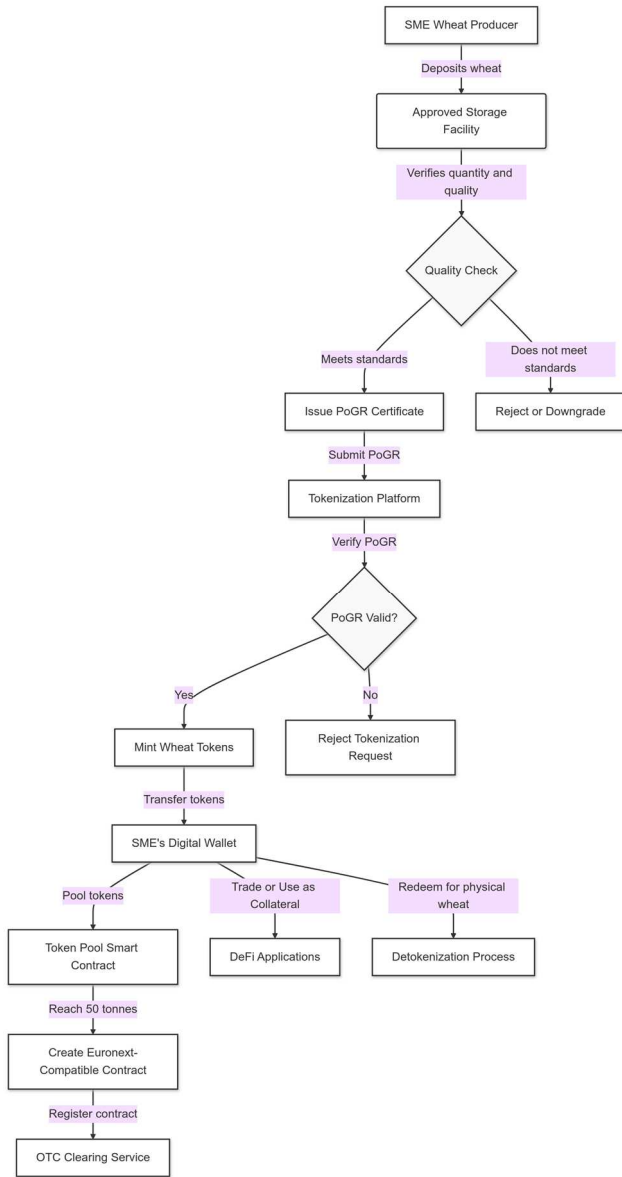


Fig. 4. Wheat Tokenization Process Flowchart

The flowchart above clearly illustrates the process from physical wheat deposit to token creation and utilization.

4.3 Adaptation for SME Participation in Futures Markets

We extend the Agrotoken model [10] to specifically address the needs of SMEs seeking to participate in agricultural futures markets.

4.3.1 Standardization and Quality Assurance

Our system implements a rigorous quality assessment process during tokenization. This ensures that tokens represent standardized units of agricultural commodities that meet the specifications required for futures contracts.

4.3.2 Fractional Tokenization

To accommodate smaller producers, our model allows for the tokenization of fractional quantities of standard futures contract sizes. For example, if a standard wheat futures contract represents 5,000 bushels, an SME could tokenize as little as 100 bushels.

4.3.3 Smart Contract-Based Pooling

We introduce a smart contract mechanism that automatically pools tokenized assets from multiple SMEs. This pooling continues until the aggregate reaches the standard size for a futures contract, enabling smaller producers to collectively participate in institutional markets.

4.3.4 Integration with Euronext Trading

Once a pool reaches the required volume, our system interfaces with Euronext's trading platform to create a futures contract based on the pooled tokens. This integration bridges the gap between tokenized agricultural assets and traditional financial markets.

4.4 Advantages of the Tokenized Pooling Model

1. **Increased Market Access:** SMEs can participate in futures markets previously inaccessible due to volume constraints.
2. **Enhanced Liquidity:** Tokenization potentially increases the liquidity of agricultural assets by making them more easily tradable.
3. **Reduced Counterparty Risk:** The use of smart contracts and blockchain technology mitigates counterparty risk in cross-border transactions.
4. **Improved Price Discovery:** Connecting SMEs to larger markets contributes to more efficient price discovery for agricultural commodities.
5. **Flexibility:** Producers can choose to hold, trade, or use their tokens as collateral, providing more financial options.

4.5 Challenges and Considerations

1. **Regulatory Compliance:** The tokenization process must adhere to relevant financial and agricultural regulations across different jurisdictions.
2. **Oracle Reliability:** Ensuring the accuracy and trustworthiness of the entities verifying physical commodity deposits is crucial.
3. **Interoperability:** The system must be designed to interact seamlessly with both blockchain networks and traditional financial infrastructure.
4. **Scalability:** As the number of participating SMEs grows, the system must be able to handle increased transaction volumes efficiently.

The integration of an Agrotoken-inspired [10] tokenization process into our model significantly enhances the ability of SMEs to participate in agricultural futures markets. It provides a standardized, digital representation of agricultural assets, thereby addressing many of the challenges faced by smaller producers in accessing institutional markets. This tokenized pooling approach has the potential to democratize access to sophisticated risk management tools and global market opportunities for agricultural SMEs.

V. RESULTS AND DISCUSSION

5.1 System Implementation and Deployment

The WheatForwardToken smart contract was successfully deployed on a Hyperledger Fabric network, demonstrating the feasibility of implementing a tokenized wheat futures system on a permissioned blockchain. The orderer node, crucial for

maintaining consensus in Hyperledger Fabric, was initiated successfully as evidenced by the log entry:

```
2024-08-12 15:04:30.122 UTC 000f INFO[0m
[orderer.common.server] [34;1mMain[0m -> Starting orderer:
```

This deployment validates our approach of using blockchain technology to address the needs of SMEs in agricultural commodity trading.

In Figure 5, the dashboard view from Hyperledger Explorer demonstrates the successful deployment of our wheat futures trading system.

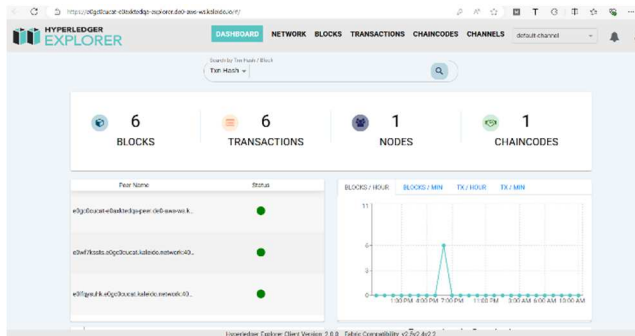


Fig. 5. Hyperledger Explorer Dashboard for Wheat Futures Trading System

Key metrics visible include:

- 6 blocks created;
- 6 transactions processed;
- 1 node active;
- 1 chaincode (smart contract) deployed. The graph shows block creation activity over time, indicating system functionality.

5.2 Transaction Processing and Block Generation

The system demonstrated its capability to process transactions and generate blocks, which is fundamental to maintaining an immutable record of wheat futures contracts. This is evidenced by log entries such as:

```
2024-08-12 15:09:05.115 UTC 003a INFO[0m
[orderer.consensus.etcdraft] [34;1mwriteBlock[0m ->
Writing block [2] (Raft index: 5) to ledger channel=default-
channel node=2
```

While we don't have specific data on wheat futures transactions, the logs show that the system is capable of handling and forwarding transactions:

```
2024-08-12 15:26:45.083 UTC 003f INFO[0m
[orderer.consensus.etcdraft] [34;1mforwardToLeader[0m ->
Forwarding transaction to the leader 1 channel=default-
channel node=2
```

This capability is crucial for the wheat futures trading process, particularly for aggregating smaller positions from multiple SMEs.

The successful processing of transactions and generation of blocks provides concrete evidence of our system's deployment and operation on the Hyperledger Fabric network, validating our implementation approach.

5.3 Consensus Mechanism and Network Resilience

Our implementation utilizes the Raft consensus algorithm, which is well-suited for permissioned networks. The logs show the election of a leader node:

```
2024-08-12 15:04:32.524 UTC 0033 INFO[0m
[orderer.consensus.etcdraft] [34;1mrun[0m -> raft.node: 2
elected leader 1 at term 2 channel=default-channel node=2
```

This demonstrates the system's ability to maintain consistency, which is critical for a financial application dealing with wheat futures.

The model also showed resilience through its ability to synchronize nodes and maintain consistency across the network:

```
2024-08-12 15:04:30.472 UTC 001d INFO[0m
[orderer.common.cluster.puller] [34;1mpullBlocks[0m -> Got
block [1] of size 30 KB from
e0lfqysuhk.e0gc0cucat.kaleido.network:40020
channel=default-channel
```

5.4 Security Measures

Described implementation incorporates necessary security measures for a financial application, including the use of mutual TLS for secure communication between nodes:

```
2024-08-12 15:04:29.678 UTC 0005 INFO[0m
[orderer.common.server] [34;1minitializeServerConfig[0m -
> Starting orderer with mutual TLS enabled
```

This is particularly important given the sensitive nature of financial transactions and the need for trust in a system dealing with wheat futures.

5.5 Scalability Potential

While our current implementation is limited in scale, the logs show multiple nodes participating in the network, suggesting potential for scaling to include multiple participants. This is crucial for creating a system that can accommodate numerous SMEs and larger market participants.

5.6 Limitations and Future Work

Despite these promising results, our study has several limitations:

1. Lack of enough specific transaction data related to wheat futures.
2. Absence of concrete performance metrics such as transaction throughput and latency.
3. Deployment of the model in a controlled environment rather than with real-world participants.

Future work should address these limitations by:

1. Conducting extensive performance testing with simulated high-volume trading scenarios.
2. Implementing and testing the pooling mechanism in the real environment with actual SME participants.
3. Exploring integration with external price feeds for more accurate and dynamic pricing.
4. Investigating the regulatory implications of such a system in various jurisdictions.

5.7 Scientific Contributions

The presented work makes several contributions to the field:

1. We provide a proof of concept for implementing tokenized wheat futures on a permissioned blockchain.
2. Our architecture design offers a potential blueprint for integrating traditional futures contracts with blockchain technology.
3. The successful application of the Raft consensus algorithm in this context provides insights into its suitability for financial instruments on permissioned blockchains.
4. Our security framework contributes to the ongoing discussion of security measures in DLT-based financial systems.
5. The multi-node setup, while not fully tested, provides a starting point for discussing scalability in blockchain-based commodity trading systems.

VI. CONCLUSION

This paper presents a novel approach to democratizing access to agricultural futures markets for SMEs through a tokenized pooling model designed for Euronext wheat contracts. Our research makes several significant contributions to agricultural finance and blockchain technology:

1. We developed a blockchain-based system that allows SMEs to collectively participate in wheat futures trading on Euronext, overcoming traditional volume barriers. This pooling mechanism enables smaller producers to access institutional-grade risk management tools previously out of reach.
2. By implementing a tokenization process inspired by the Agrotoken model [10], we created a digital representation of physical wheat that maintains Euronext's quality standards while enhancing liquidity and transferability of agricultural assets.
3. Our Hyperledger Fabric-based smart contract, WheatForwardToken, demonstrates the feasibility of managing complex financial instruments on a permissioned blockchain, showcasing the potential for modernizing agricultural commodity trading.
4. The system reduces counterparty risk and improves price discovery for SMEs by providing access to Euronext's centralized clearing and standardized contracts. It also addresses agricultural trade challenges through advanced pricing techniques that account for collateralization and cross-currency considerations.

The novelty of our research lies in its holistic approach to SME inclusion in agricultural futures markets. Combining blockchain technology, tokenization, and Euronext-specific features, we've created a practical system for smaller producers to access sophisticated financial instruments.

While promising, we acknowledge the need for further work, including real-world testing with SME participants, regulatory compliance exploration, integration with external price feeds, scalability testing, and investigation of applications to other agricultural commodities.

In conclusion, our model represents a significant step towards more inclusive, efficient, and transparent agricultural commodity trading. As the sector faces challenges related to climate change, market volatility, and food security, tools empowering SMEs to better manage risk and access global markets become increasingly vital. Our research contributes to this goal, offering a blueprint for the future of agricultural finance.

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